SPARSE MACHINE LEARNING MODEL AND PATHWAY ANALYSIS SUGGEST NOVEL GENETIC ASSOCIATIONS WITH DYSLEXIA

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Introduction

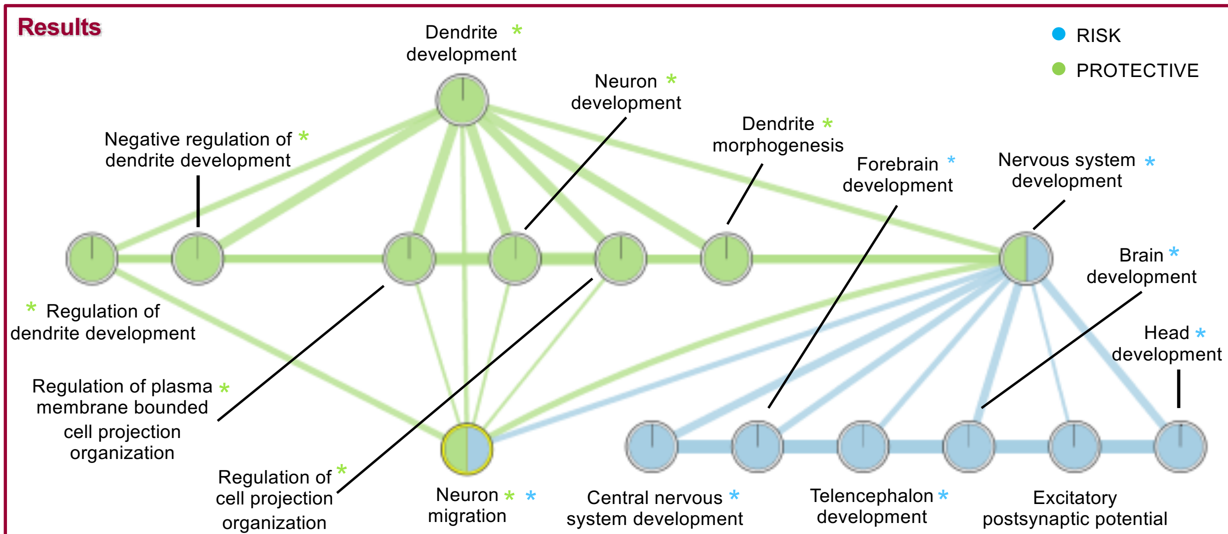
- Past genetic studies evaluated association using one-gene-at-a-time methods. Current technologies produce genetic markers in the order of **hundreds of thousands**.
- Multiple genes are associated with dyslexia.
- Genes may be located within shared biological pathway(s).
- New biological pathways are identifiable in existing population-based data sources using:
  - Advanced machine learning algorithms
  - Enrichment analysis

PURPOSE: Test multi-gene hypothesis for dyslexia and identify novel genes and pathways using sparse machine learning and enrichment analysis.

Participants

	Dyslexia	Control
Sample size	1,215	6,856
Percent male	59%	47%
Nonverbal IQ	94.56 (14.96)	102.54 (16.11)
Receptive Language	6.94 (1.75)	7.68 (1.93)
Vocabulary	8.81 (3.41)	11.83 (4.28)
Reading Comprehension	86.96 (8.36)	104.34 (9.40)
Single Word Reading	16.98 (5.41)	31.19 (7.66)

Results



Methods

Data source. The Avon Longitudinal Study of Parents And Children (ALSPAC; Boyd et al., 2013) is a prospective birth cohort. Data collection began the early 1990s and is ongoing. The sample size at year 1 was 14,684. Behavioral data collection began at age 7 with focus sessions. A selection of the original participants plus new recruits were invited to attend these sessions.

Genotyping. ALSPAC children were genotyped using the Illumina HumanHap550 quad chip genotyping platforms by 23andme. After quality control (individual call rate>0.97, SNP call rate>0.95, Minor allele frequency (MAF)>0.01, Hardy-Weinberg equilibrium (HWE) P>10-7, and removal of individuals with cryptic relatedness and non-European ancestry, 8,237 children and 477,482 SNP genotypes were retained. SNPs were flipped to the forward strand and haplotypes were estimated using ShapeIT (v2.r644). Imputation was performed using Impute V2.2.2 in 1000 genomes reference haplotypes (Version 1 Phase 3). Minor changes with respect to previously reported sample numbers are due to withdrawn consents and alterations in GWAS quality control.

Participant selection. We excluded participants with hearing loss, a diagnosis of ASD, and/or nonverbal IQ ≥ 72. Additionally, we randomly selected one twin per pair for analysis. We also identified participants who had behavioral data for all Focus sessions and those with complete behavioral data.

Phenotyping. Reading disorder status was defined as scoring less than -1 z-score on three out of ten reading measures.

Analysis:

GWAS screening → Feature selection (elastic net) → Interpretation (enrichment analysis)

Results

- **Novel genes**
  - DLC1 [chr8p22] - involved in forebrain development and negative regulation of cell migration
  - RAPGEF2 [chr4q32] - controls neuron migration and development
  - TDP1 [chr14q32.11] - plays a role in maintenance of DNA and is involved in axon development
- **Neuron migration**
  - RAPGEF2, RELN, and other dyslexia candidate genes are involved.
  - Controls several smaller pathways including the **reelin signaling pathway**, which controls positioning of neurons in the developing brain, and growth, maturation, and synaptic activity in adult brains.

Conclusions

- Results may explain why children with dyslexia show differences in neural activity compared to typical reading controls.
- Genes associated with dyslexia are over-represented in brain development pathways suggesting differences in macro-structural development.
- Potentially protective genes are overrepresented in dendrite regulation pathways suggesting differences in connections between neurons..

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For more information

https://tinyurl.com/lancaster-F32

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